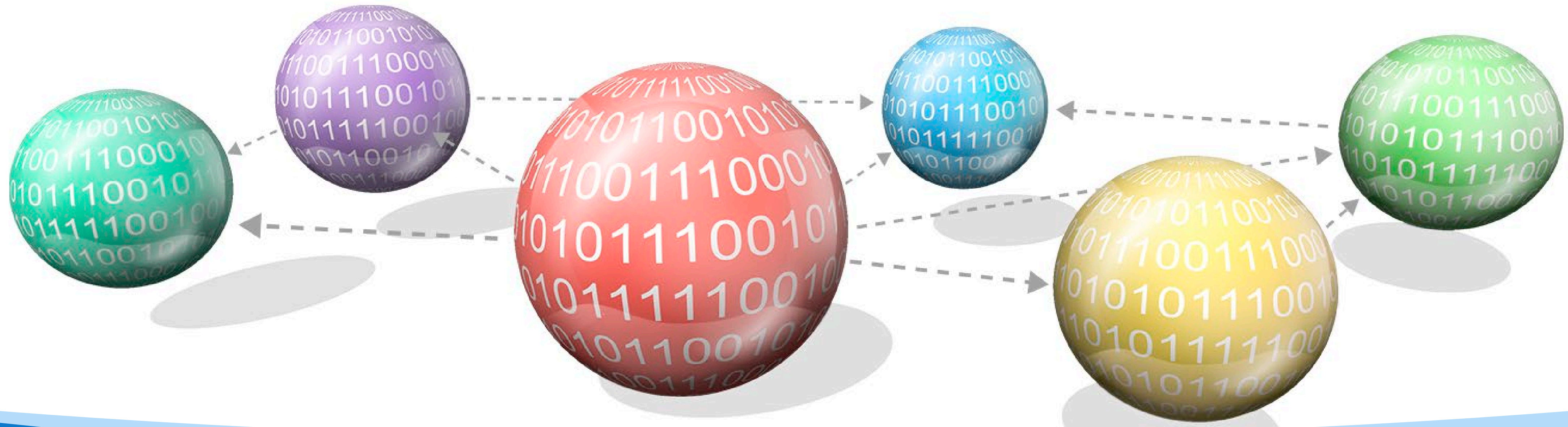

SC/CZ/CE2002 Object-Oriented Design & Programming

Assignment FAQ

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FAQ on Oct. 15

Q: May I kindly ask if there are any platforms for real-time collaboration in creating a UML diagram for the group project such as in google docs for example? To my knowledge, VP works in that changes must be individually committed to reflect them.

Ans: Yes. You can use VP online for real-time collaboration

- Student can use VP Online as their cloud base team collaboration repository.
- Please reference to the article below about how to setup the cloud repository.
- They'll need first to install the VP UML onto their laptop/PC with the license first.
- https://knowhow.visual-paradigm.com/vp-online/

Q: From our understanding, both test cases are used after the Patient sees the Doctor. Our implementation for Test Case 10 seems to completely overlap with Test Case 15.

Test Case 10: Update Patient Medical Records

- Doctor adds a new diagnosis and treatment plan to a patient's medical record.
- Verify that the medical record is updated successfully, reflecting the new information.

Test Case 15: Record Appointment Outcome

- Doctor records the outcome of a completed appointment.
- Verify that the appointment outcome is recorded, and relevant updates are visible to the patient under "View Past Appointment Outcome Records".

Ans: These are two different scenarios. The doctor can update the record without requiring an appointment.

Q: I noticed that the assignment PDF specifies, "No database application (e.g., MySQL, MS Access, etc.) is to be used." I would like to clarify whether storing and reading data from Excel files (.xlsx) or text files (.txt) is allowed for the SC2002 Health Management System (HMS) assignment.

Ans: Yes. storing and reading data from Excel files (.xlsx) or text files (.txt) are allowed.

- **No database application** (e.g., MySQL, MS Access, etc.) is to be used.
- **No JSON or XML** formats are to be used.

Q: **Do we need to strictly follow the order, properties and names of the data columns in the supplied .xlsx files or can we introduce new data columns in our own implementation?**

Ans: You can include new data columns in your own implementation.

Q: **Are we allowed to change the excel file format from .xlsx to .csv? This is because it is easier to retrieve and write data to .csv files. The project PDF only mentioned no JSON/XML format allowed.**

Ans: You are allowed to change the excel file format from .xlsx to .csv.

Q:

Is Hospital ID the same as staff ID and patient ID? Or do we need to create a running number for Hospital ID?

Ans: Hospital ID is the same as staff ID and patient ID.

Q: Are we allowed to use design patterns such as Singletons in our solution?

Ans: Yes. You can use design patterns.

Q: Can we import any external libraries to aid our solution, or are there any specific restrictions on this?

Ans: It depends on which library you want to import. The libraries for data management are not allowed.

Q: . I currently have 3 members in my project group as we are unable to find another member for our group. May I ask whether a 3-membered group is allowed and if it is, whether we will be graded more leniently as we have less members?

Ans: If your TA has made their best effort but was unable to recruit more members, we will take into consideration that your group has fewer members while assessing.

Data files

Q: Regarding CSV file usage:

- Should CSV files be used only for initial system setup (importing initial staff list, patient list, and inventory as mentioned in the requirements)?
- Or can CSV files also serve as a persistent data store throughout program execution, where we can read and write user data during runtime? This will ensure the changes to be saved even after the program terminates.

Ans: Used for both.

Q: Is it required for our project to save information between executions (data persistence)?

Ans: Yes.

Q: Is there any restrictions on the text files that we can use. For example, can we use a text file appointment.txt for the ease of scheduling appointments, as we are implementing time bounded appointments. So its hard to maintain precision, without storing the data somewhere.

So, I wanted to confirm if there is any restriction on the use of text file or do we only have to use patient.txt, staff.txt and inventory.txt?

Ans: You can have additional files.

Design

Q: Assumptions and Complex Features: What do we mean by "assumptions" in this context? For appointment scheduling, should we import time functionality, or can we assume time is not necessary for easier video demonstration?

Ans: Yes. You can assume time is not necessary for easier video demonstration and sate your assumptions.

Q: We are also thinking about adding more data to staff_list.csv - such as mentioning the doctor's specialisation. Does "assumption" in our case refer to this?

Ans: Sure. You can add column in those files.
No need to consider assumptions in file reading. If assumptions are mentioned, simply assume that the initial data can be read from the files.

Q: Regarding object-data relationships:

- Should all user details be stored completely within their respective object classes?
- Or is it acceptable to store minimal data in objects (e.g., IDs) and retrieve additional details from CSV files when needed?

Ans: It depends on how you design.

Q: Can we include like a reception class that can be like an intermediatory between all the other classes. I think this would help organise the implementations better.

Yes. You can.

Q: Can we add different implementations to how like the final bill is calculated (instead of pharmacist giving the bill, all billings will be handled by reception which will include the bill form pharmacy, consultation fee of doctor, treatment fee etc). I mean do we have to follow a strict implementation of the functions and classes given in the file. Or can we change them and include them in other classes as well.

Yes. Billing is not among the basic requirements. You can use it to demonstrate the extensibility of your system while is required.

Q:It says entity, essential control, boundary classes, and only now that I realise after talking to other groups and whatsoever that these are concepts mainly taught in SC2006 software engineering. As following the curriculum, I am not taking SC2006 this sem and was wondering how my team can approach this requirement, is there a guide for us to follow?

8. **ASSESSMENT WEIGHTAGE**

UML Class Diagram [**25 Marks**]

- Show mainly the Entity classes, the essential Control and Boundary classes, and enumeration type (if there is).

Yes. So please ignore it. As some students take 2002 and 2006 zcocurrently and keep asking me if they are allowed to put those concepts learned from 2006 into our group project. SO I state there specially for those group of students. It is allowed but not consider as basic requirements. SO I put (if there is).

Q:I thought the (if there is) is only referring to the enumeration type .

Apologies for the ambiguity. I will refine it in the next round.

Entity classes are basic requirements. Rest are optional.

Q: MVC Design Pattern: Can we confirm that we are allowed to use the MVC (Model-View-Controller) design pattern, as discussed in the last slide of Chapter 9?

Yes. You are allowed.

Q: SOLID Principles: As we are adhering to the SOLID principles, we have noticed that the number of our interfaces and classes is increasing to maintain specificity. Is this acceptable? Additionally, when justifying our application of the SOLID principles in the report, should we list every instance of usage or only a few key examples?

Yes. only a few key examples will do.

Q: Can I clarify whether we have to read/write to the original xlsx files given? Or can we just use the data and read/write using our own formats?

Ans: Yes. You can define your own format.

Q: Also does the application need to be able to persist its state even on subsequent runs?

Ans: Yes.

Q: Should the Excel file be updated every time a user updates their information (e.g. Patient updates their personal details), or are the Excel files meant for solely initializing the system?

Ans: It is up to your design.

Q: How strict is the input checking for the project? Will test cases intentionally include invalid inputs to check how the system handles them, or is the focus primarily on standard valid inputs?

Ans: Input validation is a must.

External dependencies

Q: we’re allowed to build and run our Java project using Maven. I’m concerned because our project includes a pom.xml file for dependency management, but the brief mentioned no XML files. Would this be acceptable?

Ans: Yes. You can use Maven for project management, but not for file management and tests design.

Maven

- Help developers manage and build projects efficiently.
- For file management, you have to use Java's Built-in I/O, i.e., No Maven Dependency allowed.
- For testing, you are required to write your own well-constructed rather than generic automatic template-based tests. i.e., you are required write tests that reflect realistic scenarios. For example:
  - Require tests that simulate common use cases or typical input/output flows.
  - Add test cases for edge cases that are relevant to the real-world usage of the application.

Q: Are we allowed to use external dependencies? Currently, we are using:

a. BCrypt for password salting

>>Allowed.

b. JUnit 5 for unit-testing

>>Allowed to run test but not to design test-cases.

c. Apache POI for handling Excel (.xlsx) files

>>Not allowed.

d. Miscellaneous Maven plugins for building a JAR file

>>Allowed.

Q: For SC2002 (OOP) Assignment, it was stated in FAQ that libraries are allowed except data manipulation libraries. Are we allowed to add Maven to our build and include OpenCSV for easier read/writing?

Ans: No. Additional file management library cannot be used.

To help students build a solid foundation in OODP by limiting the use of Maven

By restricting these dependencies, students will engage directly with core Java capabilities and design principles, gaining hands-on experience with the manual, foundational aspects of OODP before using advanced tools. Once they have a solid grasp of these basics, introducing these libraries can enhance their productivity without hindering their understanding.

Limited dependencies

- Basic Project Structure and Organization
- Managing JUnit for Unit Testing
- Automated Compilation and Packaging
- Building Reproducible Project Setups
- Introduction to the Build Lifecycle and Plugins

Benefit of using Maven with the restricted dependencies

With the restricted dependencies, Maven can still teach students foundational project management concepts, introduce them to automation in testing and building, and demonstrate the importance of a standardized project structure. This prepares students to adopt more advanced Maven functionalities in future projects while ensuring they understand and can implement the basics of OODP independently.

Q: I wanted to inquire about the possibility of using java.util.UUID for generating unique IDs in our group assignment. Initially, we implemented System.currentTimeMillis() and System.nanoTime() to create unique IDs for personnel. However, we've encountered occasional issues with non-unique IDs due to the efficiency of the Java runtime.
Would it be acceptable for us to use the UUID library to ensure truly unique IDs?

Ans: Yes. Can.

Q: Feature Additions: We are considering adding a feature to print bills as PDFs, which would require importing the following libraries:
[import com.itextpdf.kernel.pdf.PdfDocument; import com.itextpdf.kernel.pdf.PdfWriter; import com.itextpdf.layout.Document; import com.itextpdf.layout.element.Paragraph]

Are we permitted to use these libraries, or should we stick to the standard printing method after calculating the bill?

Yes. You can use those libraries for additional features.

Report

Q: Our team's UML diagram is quite big to fit on an A4 size page. The details might be very small or blurry. I just want to double confirm with you if we attach our UML diagram in a separate PDF file with a better resolution.

Ans: Yes. Can.

THE DELIVERABLE (Deadline:One day before your scheduled oral with your TA) Your group submission should include the following:

a. The report (separate diagram file if diagram is unclear in report)

Javadoc

Q: - What is the expected scope of Javadoc documentation - should it cover all classes, methods, and attributes, or focus on key public interfaces?.

Ans: We expect the Javadoc to cover all public classes, methods, and attributes, particularly those in key interfaces. For non-public elements, documentation can be added as needed, but the primary focus should be on public interfaces that users or other developers will interact with.

Q: What is the purpose of the Javadoc?

Ans: The Javadoc should serve as a comprehensive reference for anyone using or maintaining the code. It should provide clear descriptions of each class, method, and attribute, including any relevant details on usage, parameters, return values, and exceptions.

Q: Could you confirm the preferred method for generating and submitting the Javadoc HTML documentation?

Ans: Please use the standard Javadoc tool to generate the HTML documentation. Once generated, submit the documentation as part of the project's deliverables, either by including the HTML files in the project repository or by providing a zip file with all documentation files.

Q: Javadoc Deliverable: For the Javadoc submission, are there any specific files that we need to include? The Javadoc tool generates a complex structure with multiple HTML files, so I wanted to confirm if we should submit the whole thing or simply zip up the entire Javadoc output for submission

.

Ans: Please submit the entire Javadoc output, including all generated HTML files. The simplest approach would be to zip them up.

Q: Can what is required for submission? Is it just the modification of the code to include javadocs? Or are we required to submit the index.html file..

Ans: submit the index.html file.

Q: Just to clarify does all interfaces and classes need their respective index.html files

Ans: When you run the Javadoc tool to generate documentation for a Java project, it will create a single index.html file, which serves as the main entry point for the documentation. This main index.html page provides links to all packages, classes, and interfaces documented in the project.

Q: Testing Requirements: Regarding the testing component, could you clarify what kind of testing is expected? The brief mentions creating new test cases, and I would like to confirm if these should follow the format of the provided examples, or if we are also expected to apply white-box and black-box testing techniques..

Ans: Please follow the format of the provided examples when creating new test cases.
You are also expected to apply white-box and black-box testing techniques while developing. For demo purpose, just based on functionality.

Assessment

Q: Grading Criteria: Regarding grading, how many additional features are necessary for a high grade? If we can justify the inclusion of extra features to enhance the convenience of our HMS (Hospital Management System), while keeping the implementation at a moderate-easy level, will that affect our grading? Does adding complex features improve our grade? Based on what we understood was that designing principles is more important.

Ans: How you apply SOLID in adding complex features is important. Yes. "inclusion of extra features to enhance the convenience of our HMS (Hospital Management System), while keeping the implementation at a moderate-easy level" will improve your grading.

Group Assignment 40%
Deadline: <u>**Friday of Week 14 November 22 2024**</u>

- Group Assignment code, class diagram, and report submission deadline: One day before your scheduled oral with your TA
- Group Assignment oral presentation deadline
  - Via Physical/Online meeting
  - Arrange Group Assignment oral presentation date and time earlier with your TA
  - Every student must take turn to present

OOP Group Assignment Deliverables

- ☐ Implementation Code and Java Documentation (javadoc)
- ☐ Main Report (separate diagram file if diagram is unclear in the report)
- ☐ Data Files
- ☐ Declaration of Original Work form
- ☐ Declaration on use of GAI form
- ☐ Other Relevant files (eg: setup instruction, etc)

Students must compile ALL the required files from the checklist into <u>a single zip file</u> for submission.
File naming convention:**LabGroupName_AssignmentGroupNumber.zip**.

Upload this zip file to your individual SC/CE/CZ2002 LAB site (eg FEP1, FSP1, etc) in NTULearn. The submission link is provided on the left panel "Assignment Submission". **Only one submission is required per group.**

Peer evaluation deadline

- **Friday of Week 14 November 22 2024**
- The peer review/evaluation is a mandatory and confidential individual task. Please provide justifications for your review scores.
- Failure to complete the peer evaluation will result in a **10%** penalty on the group assignment grade.

WBS and peer evaluation

Q: Some member is inactive.

Ans: Please fill in the WBX.xls and email me and your TA.
Justify your score in peer evaluation.

[**Optional**] Member's work contribution and distribution breakdown.
If your group feels that marks should be given based on contribution, your group can fill up the WBS.xls(in the same folder as assignment doc) and include it in this report. *All members MUST consent to the WBS contents.* ***You must also email the WBS.xls to the course-coordinator with ALL members in the loop.***

Q: Where to find the TA's contact information.

Ans: NTULearn→Information.


Q&A